4. A 150 W lamp emits light of mean wavelength of 5500 Å . If the efficiency is 12%, find out the number of photons emitted by the lamp in one second.

[Ans: 4.98×10^{19}]

5. How many photons of frequency 10^{14} Hz will make up 19.86 J of energy?

[Ans: 3×10^{20}]

6. What should be the velocity of the electron so that its momentum equals that of 4000 Å wavelength photon.

[Ans: $1818\,ms^{-1}$]

7. When a light of frequency 9×10^{14} Hz is incident on a metal surface, photoelectrons are emitted with a maximum speed of $8\times10^{5}\,ms^{-1}$. Determine the threshold frequency of the surface.

[Ans: 4.61×10^{14} Hz]

8. When a 6000 Å light falls on the cathode of a photo cell, photoemission takes place. If a potential of 0.8 V is required to stop emission of electron, then determine the (i) frequency of the light (ii) energy of the incident photon (iii) work function of the cathode material (iv) threshold frequency and (v) net energy of the electron after it leaves the surface.

[Ans: 5×10^{14} Hz; 2.07 eV; 1.27 eV; 3.07×10^{14} Hz; 0.8 eV]

9. A 3310 Å photon liberates an electron from a material with energy 3×10^{-19} J while another 5000 Å photon ejects an electron with energy 0.972×10^{-19} J from the same material. Determine the value of Planck's constant and the threshold wavelength of the material.

[Ans: 6.62×10^{-34} Js; 6620×10^{-10} m]

10. At the given point of time, the earth receives energy from sun at $4\,cal\,cm^{-2}\,min^{-1}$. Determine the number of photons received on the surface of the Earth per cm^2 per minute. (Given : Mean wavelength of sun light $= 5500$ Å)

[Ans: 4.65×10^{19}]

11. UV light of wavelength 1800Å is incident on a lithium surface whose threshold wavelength is 4965Å. Determine the maximum energy of the electron emitted.

[Ans: 4.40 eV]

12. Calculate the de Broglie wavelength of a proton whose kinetic energy is equal to 81.9×10^{-15} J. (Given: mass of proton is 1836 times that of electron).

[Ans: 4×10^{-14} m]

13. A deuteron and an alpha particle are accelerated with the same potential. Which one of the two has i) greater value of de Broglie wavelength associated with it and ii) less kinetic energy? Explain.

[Ans: $\lambda_d = 2\lambda_\alpha$ and $K_d = K_\alpha/2$]

14. An electron is accelerated through a potential difference of 81 *V*. What is the de Broglie wavelength associated with it? To which part of electromagnetic spectrum does this wavelength correspond?

[Ans: $\lambda = 1.36$ Å and x-rays]

15. The ratio between the de Broglie wavelength associated with proton, accelerated through a potential of 512 V and that of alpha particle accelerated through a potential of *X* volts is found to be one. Find the value of *X*.

[Ans: 64 V]

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